

M002 Extension — Advanced Worksheet

Topic: Redstone Trail Solver (3D Cube Maze) · **Course:** Maze Madness · **Level:** Extension (Advanced) ·

Time: about 60 minutes

This is an **extension challenge** for students who have finished the weekly mazes and want something harder. The agent solves a **3D cube maze** on its own by following a **redstone trail** — reading redstone above, below, left, right and ahead, then deciding what to do at every step.

You drive it with **chat commands**: type **l** to turn left, **r** to turn right, **run** to start the solver, and **rl** to teleport the agent back to you.

1 · Read the Chat Commands

Each block of code binds a **chat word** to an action. When you type that word in chat, the agent runs the code.

```
def on_on_chat():
    agent.turn(LEFT_TURN)
    player.on_chat("l", on_on_chat)
```

What does the agent do when you type **l** in chat?


```
def on_on_chat4():
    agent.teleport_to_player()
    player.on_chat("rl", on_on_chat4)
```

You typed **rl**. Where does the agent go? When would that be useful?

Why is it handy to bind small helpers like `l`, `r` and `rl` to chat *before* you write the big `run` solver?

2 · Trace the Solver 🔍

This is the loop that runs when you type `run`. It checks redstone in different directions and reacts.

```
while True:
    agent.move(FORWARD, 1)
    if agent.detect(REDSTONE, DOWN) and agent.detect(REDSTONE, RIGHT):
        agent.move(FORWARD, 1)
        agent.interact(LEFT)
    elif agent.detect(BLOCK, RIGHT) and agent.detect(REDSTONE, LEFT):
        agent.move(FORWARD, 1)
        agent.turn(LEFT_TURN)
        agent.move(FORWARD, 1)
    elif agent.detect(REDSTONE, DOWN):
        agent.move(UP, 1)
    elif agent.detect(REDSTONE, UP):
        agent.move(DOWN, 4)
```

The loop starts with `while True`. When does it stop on its own?

The redstone is a *trail* the agent follows. In your own words, what does `detect(REDSTONE, DOWN)` tell the agent?

This maze is 3D — the agent goes up and down, not just flat. Which two lines move it between floors? What does each one do?

Why does the order of the `if` / `elif` checks matter? What would change if the `elif agent.detect(REDSTONE, DOWN)` check came *first*, before the others?

3 · Spot the Bug 🐛

Each block below is broken. Read what it is **supposed** to do, rewrite it so it works, then explain the fix.

Bug A — The solver should keep going until the maze is done, checking the trail on every step.

```
agent.move(FORWARD, 1)
if agent.detect(REDSTONE, DOWN):
    agent.move(UP, 1)
```

Hint: this only checks *once*. What wraps a block so it repeats?

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — When there is redstone **down and to the right**, the agent should step forward then flip a lever on its **left**.

```
if agent.detect(REDSTONE, DOWN) or agent.detect(REDSTONE, RIGHT):
    agent.interact(LEFT)
```

Hint: "down **and** right" — both must be true. And it should move forward before it interacts.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Make It Your Own

Open the extension world and run the solver with `run`. Now **change one thing** and predict what happens before you test it.

Pick **one** idea (or your own):

- Add a new chat command, like `back`, that turns the agent around (two right turns).
- Change `agent.move(DOWN, 4)` to a different number and watch where the agent lands.
- Add an `elif` that does something when there is redstone straight **ahead**.

Which change did you pick? Write the code for it.

Predict first: what do you think will happen when you run it?

Then test it. What actually happened? If it surprised you, why?

5 · Tell Me What You Built 📷

When the agent finishes the cube maze, come back here. Send a photo or video of the agent reaching the goal, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

The maze was 3D, so the agent had to ...

The redstone trail told the agent to ...

I used `detect` to ...

The change I made in section 4 was ...

The hardest part was ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while the agent solves the maze. Talk through it like you are teaching someone who has never seen it. Try to use these words: **detect**, **redstone**, **trail**, **loop**, **3D**, **elif**.

1. Show the cube maze and the redstone trail before you start.
2. Type `run` and show the agent following the trail.
3. Read one `if` / `elif` line out loud and say what decision it makes.
4. Show the change you made in section 4 and what it did.

Write what you will say in your video. Plan it here first — you can read from it while filming.

Submit ☒

Send this worksheet + your walkthrough video to teacher on KakaoTalk.